

SINGLE-PHASE GAS FLOW IN MICROCHANNELS

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2.2. Gas flow regimes in microchannels

Usually, the **regime** of gas flow in a **macrochannel** refers to; **viscous** and **compressibility** effects, respectively **quantified** by the Reynolds number **Re** and the Mach number **Ma**.

Typically for $Re < 2000 \rightarrow \text{laminar flow}$
 $Re > 2000 \rightarrow \text{turbulent flow}$

$Ma < 1 \rightarrow \text{subsonic flow}$
 $Ma > 1 \rightarrow \text{supersonic flow}$

Typically, if **Ma** > 0.3, **compressibility** effects should be taken into account.

In addition to **Re** and **Ma**, in **microchannels** **Rarefaction effects** $\rightarrow$ main effects (quantified by **Kn**)

Consequently, the **regime** usually refers to **Re**, **Ma** and **Kn** effects, according to the classification given in Section 2.1.3, and flows are considered as **microflows** for **Kn** > 10^{-3} .

Actually, Eq. (2.22) shows that for a dilute gas,

$$Re = k_2 \sqrt{\gamma} Ma / Kn$$

$k_2 \sqrt{\gamma}$ is of the order of **unity**, which implies that; for **subsonic** microflows, **viscous effects** are predominant and the flows are **laminar**. (Re number is not high-LK)

The models used to describe gas flows in microchannels depend on the rarefaction regime. The main **models** are presented in **Fig. 2.7**, with the rough values of **Kn**.

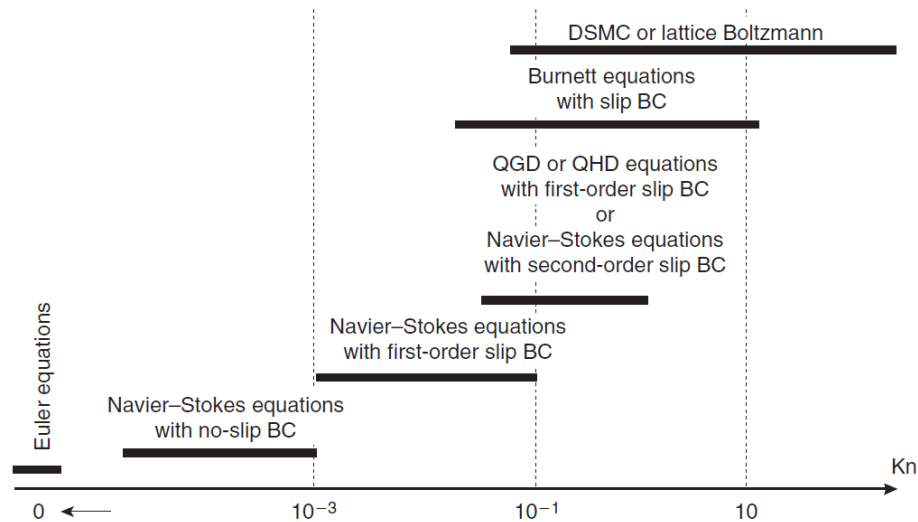


Fig. 2.7. Gas flow regimes and main models according to the Knudsen number. BC: boundary condition. (quasi-hydrodynamic-QHD, quasi-gasdynamic-QGD - LK)

The limits are **indicative** and may vary according to the considered configurations.

These models are described in the following sections.

Among the three main rarefied regimes

(**slip flow**, **transition flow** and **molecular flow**),

the **slip flow** regime is more specially detailed, since

it is the **most frequently encountered** (cf. **Fig. 2.6**) and

it allows **analytic** or **semi-analytic** developments.

Moreover, the gas is considered as a **dilute gas**, which makes valid the **ideal gas** assumption.

2.2.1. Ideal gas model

The classic equation of state for an ideal gas,

$$p = \rho RT = nk_B T \quad (2.24)$$

is also called the Boyle–Mariotte's law. (Denklemin sağ tarafi incelensin-LK)

p is the pressure,

$\rho = mn$ is the density,

$k_B = Ru/N = (M/N)(Ru/M) = mR = 1.38065 \text{ J/K}$ is the Boltzmann constant,

$Ru = 8.314511 \text{ J/mol/K}$ being the universal gas constant,

$N = 6.022137 \times 10^{23} \text{ mol}^{-1}$, the Avogadro's number and
 M , the molecular weight.

Equation (2.24) may be obtained from a molecular analysis, assuming that the gas is a dilute gas in thermodynamic equilibrium.

If the thermodynamic equilibrium is not verified,

Eq. (2.24) (ideal gas eq.-LK) remains valid, providing the

thermodynamic temperature T

is replaced with the

translational kinetic temperature T_{tr} ,

which measures the mean translational kinetic energy of the molecules.

The specific ideal gas constant

$$R = c_p - c_v \quad (2.25)$$

is the difference between the specific heats

$$c_p = \left. \frac{\partial h}{\partial T} \right|_p \quad (2.26)$$

and

$$c_v = \left. \frac{\partial e}{\partial T} \right|_v \quad (e = u \quad \text{LK}) \quad (2.27)$$

where e is the internal specific energy, $v = 1/\rho$ is the specific volume and h is the specific enthalpy.

For an ideal gas, the specific heats are constant and their ratio is the index

$$\gamma = \frac{c_p}{c_v} \quad (\gamma = k \quad \text{LK}) \quad (2.28)$$

Therefore, for an ideal gas,

$$c_v = \frac{1}{\gamma - 1} R \quad c_p = \frac{\gamma}{\gamma - 1} R \quad (2.29)$$

The specific heat ratio is constant for a monatomic gas ($\gamma = 5/3$)

but it is dependent of the temperature for a diatomic gas, with a value close to 7/5 near atmospheric conditions.

Finally, we can note that for an ideal gas and an IPL collision model, the Knudsen number can be written as:

$$\lambda = k_2 \mu / \rho \sqrt{RT} \quad p = \rho RT$$

$$\text{Kn} = \frac{\lambda}{L} \quad \text{Kn} = \frac{k_2 \mu(T)}{\rho \sqrt{RT} L} = \frac{k_2 \mu(T) \sqrt{RT}}{pL} \quad (2.30)$$

with the value of k_2 given in Table 2.1.

2.2.2. Continuum flow regime $\text{Kn} < 10^{-3}$

The classic continuum flow regime (which corresponds to the shaded area in Fig. 2.5) may be accurately modeled by;

the compressible Navier–Stokes equations,

the ideal gas equation of state, and

classic boundary conditions

that express the continuity of temperature and velocity between the fluid and the wall.

2.2.2.1. Compressible Navier–Stokes equations

The compressible Navier–Stokes equations are the governing conservation laws for

mass,

momentum and

energy

(Navier–Stokes equations refers only to conservation of momentum – LK)

These laws are written assuming that the fluid is Newtonian, so that the stress tensor

$$\boldsymbol{\sigma}' = -p\mathbf{I} + \boldsymbol{\sigma} \quad (2.31)$$

is a linear function of the velocity gradients. Thus, the viscous stress tensor has the form

$$\boldsymbol{\sigma} = \mu[\nabla \otimes \mathbf{u} + (\nabla \otimes \mathbf{u})^T] + \zeta(\nabla \cdot \mathbf{u})\mathbf{I}$$

where $\mathbf{u}$ is the velocity vector, $\mathbf{I}$ is the unit tensor, μ is the dynamic viscosity and ζ is the second coefficient of viscosity, or Lamé coefficient.

With Stokes' hypothesis, $2\mu + 3\zeta = 0$, which expresses that the changes of volume do not involve viscosity, the viscous stress tensor may be written as

$$\boldsymbol{\sigma} = \mu[\nabla \otimes \mathbf{u} + (\nabla \otimes \mathbf{u})^T - (2/3)(\nabla \cdot \mathbf{u})\mathbf{I}]$$

The validity of this assumption is discussed in Gad-el-Hak (1995).

The Navier–Stokes equations also assume that the fluid follows the Fourier law of diffusion. Thus, the heat flux vector $\mathbf{q}$ is related to the temperature gradient by $\mathbf{q} = -k\nabla T$, where the thermal conductivity k is a function of the temperature.

With the above relations, the compressible Navier–Stokes equations may be written as:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (2.32)$$

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot \left(\rho \mathbf{u} \otimes \mathbf{u} - \mu \left[\nabla \otimes \mathbf{u} + (\nabla \otimes \mathbf{u})^T - \frac{2}{3}(\nabla \cdot \mathbf{u})\mathbf{I} \right] \right) + \nabla p = \mathbf{f} \quad (2.33)$$

$$\frac{\partial E}{\partial t} + \nabla \cdot \left((E + p)\mathbf{u} - \mu \left[\nabla \otimes \mathbf{u} + (\nabla \otimes \mathbf{u})^T - \frac{2}{3}(\nabla \cdot \mathbf{u})\mathbf{I} \right] \cdot \mathbf{u} - k \nabla T \right) = \mathbf{f} \cdot \mathbf{u} \quad (2.34)$$

These conservation equations are respectively;

the continuity eq. (2.32),

the momentum eq. (2.33)

the energy eq. (2.34)

The total energy per unit volume

$$E = \rho \left(\frac{1}{2} \mathbf{u} \cdot \mathbf{u} + e \right) \quad (2.35)$$

is a function of the internal specific energy e that may be written for an ideal gas, using Eqs. (2.24)–(2.28)

$$e = \frac{p}{\rho(\gamma - 1)} \quad (2.36)$$

The external volume forces $\mathbf{f}$ (gravitational forces, magnetic forces, etc.) are generally

negligible in gas flow.

It is even more the case in gas microflows,

since the volume over surface ratio decreases with the characteristic length L and consequently, in microscale geometries,

volume effects

may be neglected when compared to

surface effects.

2.2.2.2. Classic boundary conditions

The previous set of Eqs.

Ideal Gas Eq. (2.24)

Conservation of mass (2-32)

Conservation of momentum (2-33)

Conservation of energy (2-34)

must be completed with appropriate boundary conditions.

In the **continuum flow regime**, they express

the continuity of the velocity,

$$\mathbf{u}|_w = \mathbf{u}_{\text{wall}} \quad (2.37)$$

and, the continuity of the temperature,

$$T|_w = T_{\text{wall}} \quad (2.38)$$

at the wall,

the subscripts “wall” relating to the wall itself, and

“w” to the conditions in the fluid at the wall.

2.2.3. Slip flow regime

$$10^{-3} < \text{Kn} < 10^{-1}$$

Actually, whatever the Knudsen number (even for very low values of Kn), there is in the neighborhood of the wall a domain in which the gas is out of equilibrium.

This domain is called the **Knudsen layer** and has a thickness in the order of the mean free path.

For very low Knudsen numbers (in the continuum flow regime), the effect of the Knudsen layer is negligible.

In the slip flow regime, which is roughly in the range $10^{-3} < \text{Kn} < 10^{-1}$, the Knudsen layer must be taken into account.

Actually, the flow in the Knudsen layer cannot be analyzed from continuum by use of the Navier–Stokes equations, for example.

But for $\text{Kn} < 10^{-1}$ (slip flow regime-LK), its thickness is small enough and the Knudsen layer can be neglected,

providing

the boundary conditions ($\mathbf{u}|_w = \mathbf{u}_{\text{wall}}$, $T|_w = T_{\text{wall}}$) are modified and express a velocity slip,

as well as a

temperature jump

at the wall (Section 2.2.3.2).

Thus, the Navier–Stokes equations remain applicable,

but it could also be convenient to replace them by another set of conservation equations, such as;

the quasi-hydrodynamic (QHD) equations or
the quasi-gasdynamics (QGD) equations (see Section 2.2.3.1).

2.2.3.1. Continuum NS–QGD–QHD equations

In a **generic form**, the conservation equations may be written in the following way:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J}_{\text{EQ}} = 0 \quad (2.39)$$

for the **continuity** equation,

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\mathbf{J}_{\text{EQ}} \otimes \mathbf{u} - \boldsymbol{\sigma}_{\text{EQ}}) + \nabla p = 0 \quad (2.40)$$

for the **momentum** equation and

$$\frac{\partial E}{\partial t} + \nabla \cdot \left(\frac{E + p}{\rho} \mathbf{J}_{\text{EQ}} - \boldsymbol{\sigma}_{\text{EQ}} \cdot \mathbf{u} + \mathbf{q}_{\text{EQ}} \right) = 0 \quad (2.41)$$

for the **energy** equation.

In these equations;

$\mathbf{J}_{\text{EQ}}$ is the mass flux vector,

$\boldsymbol{\sigma}_{\text{EQ}}$, the viscous stress tensor and

$\mathbf{q}_{\text{EQ}}$, the heat flux vector.

The external volume forces $\mathbf{f}$ have been neglected.

In the case of the **Navier–Stokes equations** (EQ $\equiv$ NS) these variables have the form:

$$\mathbf{J}_{\text{NS}} = \rho \mathbf{u} \quad (2.42)$$

$$\boldsymbol{\sigma}_{\text{NS}} = \mu [\nabla \otimes \mathbf{u} + (\nabla \otimes \mathbf{u})^T] + \zeta (\nabla \cdot \mathbf{u}) \mathbf{I} = \mu \left[\nabla \otimes \mathbf{u} + (\nabla \otimes \mathbf{u})^T - \frac{2}{3} (\nabla \cdot \mathbf{u}) \mathbf{I} \right] \quad (2.43)$$

$$\mathbf{q}_{\text{NS}} = -k \nabla T \quad (2.44)$$

These lead to the previous Eqs. (2.32)–(2.34).

In the Navier–Stokes equations, the density ρ , the pressure p and the velocity $\mathbf{u}$ correspond to space averaged instantaneous quantities.

From spatiotemporal considerations, the QGD and the QHD equations may be obtained (Lengrand and Elizarova, 2004). They are based on the same governing conservation laws (2.39)–(2.41), but these laws are closed with a different definition of $\mathbf{J}_{\text{EQ}}$, $\boldsymbol{\sigma}_{\text{EQ}}$ and $\mathbf{q}_{\text{EQ}}$. Dissipative terms are added, which involve a parameter τ . This parameter

$$\tau = \frac{\gamma\mu}{Sc \rho a^2} \quad (2.45)$$

is a time characteristic of the temporal averaging. Here, $Sc=\mu/(\rho D)$ is the Schmidt number, which represents the ratio of the diffusive mass over the diffusive momentum fluxes. In the expression of Sc , D is the diffusion coefficient, which can be expressed as a function of α and ω for a VSS molecule. Thus, for a VSS molecule,

$$Sc = \frac{5(2 + \alpha)}{3(7 - 2\omega)\alpha} \quad (2.46)$$

which reduces to 5/6 for a HS model. For an ideal gas, $a^2 = \gamma RT$ and τ may be written as:

$$\tau = \frac{1}{Sc} \frac{\mu}{p} \quad (2.47)$$

The QGD equations are appropriate for dilute gases. For these equations (EQ≡QGD):

$$\mathbf{J}_{\text{QGD}} = \mathbf{J}_{\text{NS}} - \tau[\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) + \nabla p] \quad (2.48)$$

$$\boldsymbol{\sigma}_{\text{QGD}} = \boldsymbol{\sigma}_{\text{NS}} + \tau[\mathbf{u} \otimes (\rho \mathbf{u} \cdot (\nabla \otimes \mathbf{u}) + \nabla p) + \mathbf{I}(\mathbf{u} \cdot \nabla p + \gamma p \nabla \cdot \mathbf{u})] \quad (2.49)$$

$$\mathbf{q}_{\text{QGD}} = \mathbf{q}_{\text{NS}} - \tau \rho \mathbf{u}[\mathbf{u} \cdot \nabla e + p \mathbf{u} \cdot \nabla(1/\rho)] \quad (2.50)$$

From Eqs. (2.18), (2.19), (2.24) and (2.47), we can deduce that for a dilute gas and an IPL collision model,

$$\tau = \frac{Kn}{Sc} \frac{1}{k_2} \frac{L}{\sqrt{RT}} \quad (2.51)$$

Therefore, $[\tau \nabla]_{(Kn \rightarrow 0)} \rightarrow 0$, which shows that the above QGD equations degenerate into the classic NS equations when rarefaction is negligible.

Another set of equations, the QHD equations, are appropriate to dense gases. In that case (EQ=QHD):

$$\mathbf{J}_{\text{QHD}} = \mathbf{J}_{\text{NS}} - \tau[\rho \mathbf{u} \cdot (\nabla \otimes \mathbf{u}) + \nabla p] \quad (2.52)$$

$$\sigma_{\text{QHD}} = \sigma_{\text{NS}} + \tau \mathbf{u} \otimes (\rho \mathbf{u} \cdot (\nabla \otimes \mathbf{u}) + \nabla p) \quad (2.53)$$

$$\mathbf{q}_{\text{QHD}} = \mathbf{q}_{\text{NS}} \quad (2.54)$$

They also bring additional terms to the NS equations, which are proportional to the parameter τ .

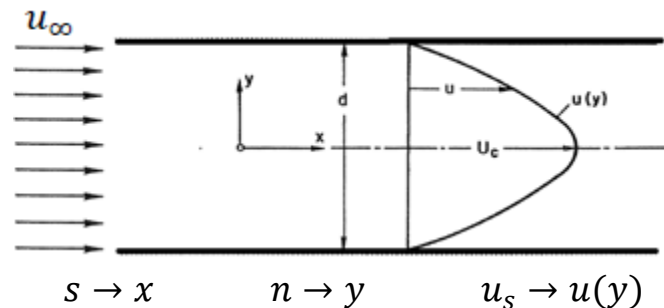
2.2.3.2. First-order slip boundary conditions

From a series of experiments at low pressures performed in 1875, Kundt and Warburg were probably the first who pointed out that for a flow of rarefied gas, slipping occurs at the walls (Kennard, 1938).

For a gas flowing in the direction s parallel to the wall, the slip velocity may be written as:

$$u_{\text{slip}} = u_s - u_{\text{wall}} = \xi \left. \frac{\partial u_s}{\partial n} \right|_w \quad (2.55)$$

where ξ is a length commonly called the **coefficient of slip** and n is the normal direction, exiting the wall. (u_s fluid velocity – LK)



With appropriate assumptions, it is possible to calculate the magnitude of the coefficient of slip from the kinetic theory of gases.

Following a paper initially dealing with stresses in rarefied gases due to temperature gradients (Maxwell, 1879), Maxwell added an appendix on the advise of a referee, with the aim of expressing the conditions which must be satisfied by a gas in contact with a solid body.

Maxwell treated the surface as something intermediate between a perfectly reflecting and a perfectly absorbing surface.

Therefore, he assumed that for every unit of area,

a fraction σ of the molecules is absorbed by the surface

(due to the roughness of the wall, or to a condensation–evaporation process (Kennard, 1938))

and is afterwards reemitted with

Durgun gaz velocities corresponding to those in **still gas** at the **temperature of the wall**.

The other **fraction, $1 - \sigma$** , of the molecules is **perfectly reflected** by the wall (Fig. 2.8).

The dimensionless coefficient **σ** is called the

tangential momentum accommodation coefficient.

When **$\sigma = 0$** , the **tangential momentum** of the **incident molecules** **equals** that of the **reflected molecules** and no momentum is transmitted to the wall, as if the flow was inviscid. This kind of reflection is called **specular reflection**.

Conversely, when **$\sigma = 1$** , the **gas molecules** **transmit** **all their tangential momentum**

to the **wall** and the reflection is a **diffuse reflection**.

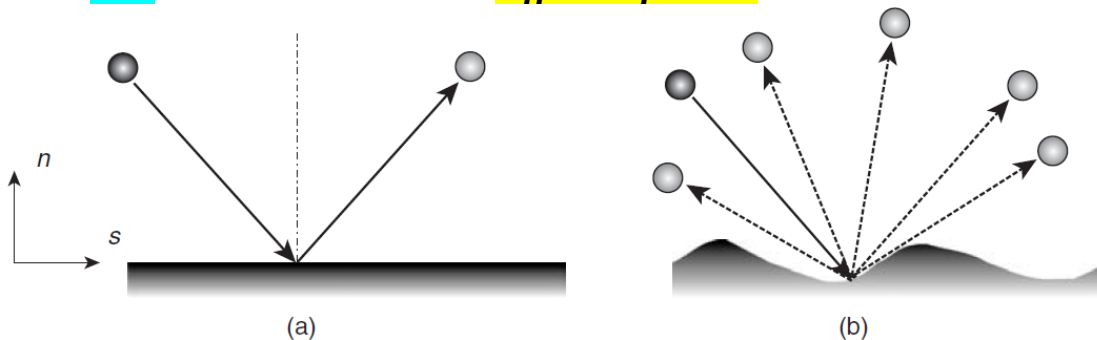


Fig. 2.8. Maxwell hypothesis. (a) Perfect (specular) reflection of a fraction $1 - \sigma$ of the molecules. (b) Absorption and diffuse reemission of a fraction σ of the molecules.

From a **momentum balance at the wall**, Maxwell finally demonstrated that there is a **slip at the wall**, which takes the form:

$$u_{\text{slip}} = u_s - u_{\text{wall}} = \frac{2 - \sigma}{\sigma} \lambda_M \left[\frac{\partial u_s}{\partial n} - \underbrace{\frac{3}{2} \frac{\mu}{\rho T} \frac{\partial^2 T}{\partial s \partial n}}_{\text{Neglect}} \right]_w + \frac{3}{4} \left[\frac{\mu}{\rho T} \frac{\partial T}{\partial s} \right]_w \quad (2.56)$$

with

$$\lambda_M = \sqrt{\frac{\pi}{2}} \frac{\mu}{\sqrt{p\rho}} = \sqrt{\frac{\pi}{2}} \frac{\mu}{\rho \sqrt{RT}} \quad (2.57)$$

which is **very close** to the value of the **mean free path λ_{HS}** deduced from an HS model.

The equation of Maxwell is generally cited in the literature with the form

$$u_{\text{slip}} = u_s - u_{\text{wall}} = \frac{2 - \sigma}{\sigma} \lambda \left. \frac{\partial u_s}{\partial n} \right|_w + \frac{3}{4} \frac{\mu}{\rho T} \left. \frac{\partial T}{\partial s} \right|_w$$

$$= \frac{2 - \sigma}{\sigma} \lambda \left. \frac{\partial u_s}{\partial n} \right|_w + \frac{3}{4} \frac{\lambda}{k_2} \sqrt{\frac{R}{T}} \left. \frac{\partial T}{\partial s} \right|_w \quad (2.58)$$

which omits the second term on the right-hand side of Eq. (2.56).

After non-dimensionalization with the characteristic length L , a reference velocity u_0 and a reference temperature T_0 , Eq. (2.58) is written as follows:

$$u_s^* - u_{\text{wall}}^* = \frac{2 - \sigma}{\sigma} \text{Kn} \left. \frac{\partial u_s^*}{\partial n^*} \right|_w + \frac{3}{4} \frac{\text{Kn}^2 \text{Re}_0}{\gamma k_2^2} \frac{\text{Ma}_0^2}{\text{Ma}_0^2} \left. \frac{\partial T^*}{\partial s^*} \right|_w \quad (2.59)$$

Where

$$u_s^* = u_s/u_0, T^* = T/T_0, n^* = n/L \text{ and } s^* = s/L$$

Note that this non-dimensionalization is valid only if the reference lengths are the same in the s and n directions, which is generally not the case for a flow in a microchannel.

In that case, it is preferable to keep the dimensional form of Eq. (2.58).

(For non-dimensional case Eq. 2.59 - LK) The second term on the right-hand side, which is proportional to the square of the Knudsen number $\text{Kn} = \lambda/L$, involves the Reynolds number $\text{Re}_0 = \rho u_0 L / \mu$ and the Mach number $\text{Ma}_0 = u_0 / \sqrt{\gamma R T_0}$ defined at the reference conditions. The coefficient k_2 depends on the intermolecular collision model and its value is given in Table 2.1.

This second term is associated with the

thermal creep, or transpiration phenomenon.

It shows that a flow can be caused in the sole presence of a

tangential temperature gradient, without any pressure gradient.

In that case, the gas experiences a **slip velocity** and moves from the **colder** toward the **warmer** region (Why from colder toward the warmer region? LK).

This can also result in **pressure variation** in a microchannel only submitted to **tangential temperature gradients**.

The non-dimensional form of Eq. (2.59) clearly shows that **slip** is **negligible** for **very low Knudsen numbers**, but should be taken into account when Kn is no longer small, typically as soon as it becomes higher than 10^{-3} .

In **analogy** with the **slip phenomenon**, **Poisson** suggested that there might also be a **temperature jump** at the wall, which could be described by an equation equivalent to

$$T - T_{\text{wall}} = \varsigma \left. \frac{\partial T}{\partial n} \right|_{\text{w}} \quad (2.60)$$

where ς is called the **temperature jump distance**.

Smoluchowski experimentally confirmed this hypothesis (Smoluchowski, 1898) and showed that ς was **proportional** to the **mean free path λ** .

As for the slip, it can be assumed that a **fraction σ_T** of the molecules have a **long contact** with the wall and that the **wall adjusts** **their mean thermal energy**.

Therefore, **these molecules** are **reemitted** as if they were issuing from a gas at the **temperature of the wall**.

The other **fraction, $1 - \sigma_T$** , is reflected **keeping** its **incident thermal energy**.

The dimensionless coefficient **σ_T** is called the **energy accommodation coefficient**.

From an energy balance at the wall, it can be shown that (Kennard, 1938):

$$T - T_{\text{wall}} = \frac{2 - \sigma_T}{\sigma_T} \frac{2\gamma}{\gamma + 1} \frac{k}{\mu c_v} \lambda \left. \frac{\partial T}{\partial n} \right|_{\text{w}} \quad (2.61)$$

After non-dimensionalization with the characteristic length L and the reference temperature T_0 , Eq. (2.61) is written as follows:

$$T^* - T_{\text{wall}}^* = \frac{2 - \sigma_T}{\sigma_T} \frac{2\gamma}{\gamma + 1} \frac{\text{Kn}}{\text{Pr}} \frac{\partial T^*}{\partial n^*} \Big|_{\text{w}} \quad (2.62)$$

where

$$\text{Pr} = \frac{\nu \rho c_p}{k} = \frac{\mu c_p}{k} \quad (2.63)$$

is the Prandtl number.

In the absence of tangential temperature gradients ($\partial T^*/\partial s^* = 0$), the boundary conditions (2.59) and (2.62) are called

$$u_s^* - u_{\text{wall}}^* = \frac{2 - \sigma}{\sigma} \text{Kn} \frac{\partial u_s^*}{\partial n^*} \Big|_{\text{w}}$$

$$T^* - T_{\text{wall}}^* = \frac{2 - \sigma_T}{\sigma_T} \frac{2\gamma}{\gamma + 1} \frac{\text{Kn}}{\text{Pr}} \frac{\partial T^*}{\partial n^*} \Big|_{\text{w}}$$

first-order (i.e. $\mathcal{O}(\text{Kn})$)

boundary conditions.

The velocity slip is then proportional to the transverse velocity gradient and to the Knudsen number Kn . (LK)

The temperature jump is then proportional to the transverse temperature gradient and to the Knudsen number Kn . (LK)

These boundary conditions are classically associated with the Navier–Stokes equations, but they can also lead to interesting results when used with the QGD or QHD equations.

2.2.3.3. Higher-order slip boundary conditions

From a theoretical point of view, the slip flow regime is particularly interesting because it generally leads to analytical or semi-analytical models (cf. Section 2.3).

For example, these analytical models allow us to calculate velocities and flow rates for isothermal and locally fully developed microflows between plane plates or in cylindrical ducts with simple sections: circular (Kennard, 1938), annular (Ebert and Sparrow, 1965), rectangular (Ebert and Sparrow, 1965; Morini and Spiga, 1998)...

These models proved to be quite precise for moderate Knudsen numbers, typically up to about 0.1 (Harley *et al.*, 1995; Liu *et al.*, 1995; Shih *et al.*, 1996; Arkilic *et al.*, 2001).

For $Kn > 0.1$,

experimental studies (Sreekanth, 1969) ,or,

numerical studies (Piekos and Breuer, 1996)

with the direct simulation Monte Carlo (DSMC) method

show significant deviations

with models based on

first-order boundary conditions.

Therefore, since 1947, several authors have proposed

second-order boundary conditions,

hoping to extend the validity of the

slip flow regime to higher Knudsen numbers.

Second-order boundary conditions take more or less complicated forms, which are difficult to group together in a sole equation.

Actually, according to the assumptions, the

second-order terms ($\mathcal{O}(Kn^2)$)

may be dependent of

σ (Chapman and Cowling, 1952; Karniadakis and Beskok, 2002)

and involve

tangential second derivatives $\partial^2 u^*/\partial s^{*2}$ (Deissler, 1964).
In the simple case of a developed flow between plane plates,
the tangential second derivatives
are

zero
and one may compare most of the second-order models that take the generic
form

$$u_s^* - u_{\text{wall}}^* = A_1 \text{Kn} \frac{\partial u_s^*}{\partial n^*} + A_2 \text{Kn}^2 \frac{\partial^2 u_s^*}{\partial n^{*2}} \quad (2.64)$$

In the particular case of a fully diffuse reflection ($\sigma = 1$), the coefficients A_1 and A_2 proposed in the literature are compared in Table 2.3.

We note significant differences, essentially for the second-order term.
(A_2 LK)

Moreover, some models based on a simple mathematical extension of Maxwell's condition predict

a decrease of the slip compared to the first order model,
while other models predict
an increase of the slip.

This point is discussed in Section 2.3.

Table 2.3

Coefficients of the main models of second-order boundary conditions proposed in the literature, for $\sigma = 1$.

Author, year (reference)	A_1	A_2
Maxwell (Maxwell, 1879)	1	0
Schamberg, 1947 (Karniadakis and Beskok, 2002)	1	$-5\pi/12$
Chapman and Cooling (Chapman and Cowling, 1952)	$\kappa_0 (\approx 1)$	$\kappa_0^2/2 (\approx 1/2)$
Deissler (Deissler, 1964)	1	$-9/8$
Cercignani, 1964 (Sreekanth, 1969)	1.1466	-0.9756
Hsia et Domoto (Hsia and Domoto, 1983)	1	$-1/2$
Mitsuya (Mitsuya, 1993)	1	$-2/9$
Karniadakis et Beskok (Karniadakis and Beskok, 2002)	1	$1/2$

The implementation of the boundary conditions (2.64)

$$u_s^* - u_{\text{wall}}^* = A_1 \text{Kn} \frac{\partial u_s^*}{\partial n^*} + A_2 \text{Kn}^2 \frac{\partial^2 u_s^*}{\partial n^{*2}} \quad (2.64)$$

with the Navier–Stokes equations

is not always easy

and can lead to computational difficulties.

The obvious interest in using higher-order boundary conditions led some authors to propose new conditions, which may be

more appropriate

for a treatment with the Navier–Stokes equations. Thus, Beskok and Karniadakis (1999) suggested a high-order form

$$u_s^* - u_{\text{wall}}^* = \frac{2 - \sigma}{\sigma} \frac{\text{Kn}}{1 - A_3 \text{Kn}} \frac{\partial u_s^*}{\partial n^*} \quad (2.65)$$

that involves the sole first derivative of the velocity and an empirical parameter $A_3(\text{Kn})$. As for Xue and Fan (2000), they proposed

$$u_s^* - u_{\text{wall}}^* = \frac{2 - \sigma}{\sigma} \tanh(\text{Kn}) \frac{\partial u_s^*}{\partial n^*} \quad (2.66)$$

that leads to results close to that calculated by the DSCM method, up to high Knudsen numbers, of the order of 3. Other hybrid boundary conditions, such as

$$u_s^* - u_{\text{wall}}^* = \frac{2 - \sigma}{\sigma} \left[\text{Kn} \frac{\partial u_s^*}{\partial n^*} + \frac{\text{Kn}}{2} \left(\text{Re} \frac{\partial p^*}{\partial s^*} \right) \right] \quad (2.67)$$

Jie *et al.* (2000), were also proposed in order to allow a more stable numerical solution, while giving results comparable to that from Eq. (2.65) for the cases that were tested.

2.2.3.4. Accommodation coefficients

The accommodation coefficients depend on various parameters that affect surface interaction, such as the magnitude and the direction of the velocity.

Fortunately, it seems that these coefficients are reasonably constant for a given gas and surface combination (Schaaf, 1963). Many measurements of σ_T and σ have been made, most of them from indirect macroscopic measurements, while a few of them are from direct measurements using molecular beam techniques.

Schaaf (1963) reported values of σ ranging from 0.60 to 1.00, most of them being between 0.85 and 1.00.

Recently, good correlations for microchannel flows were found in the slip flow regime, with values of σ between 0.80 and 1.00 (Section 2.3.5.2).

Schaaf also reported values of σ_T in a wide range, from 0.0109 to 0.990.

2.2.4. Transition flow and free molecular flow

Typically, for Knudsen numbers higher than unity, the continuum Navier–Stokes or QGD/QHD equations become invalid.

However, in the early transition regime (for $Kn \sim 1$), a slip-flow model may still be used providing the continuum model is corrected by replacing the Navier–Stokes equations with the Burnett equations.

For higher Knudsen numbers, in the full transition and in the free molecular regimes, the Boltzmann equation must be directly treated by appropriate numerical techniques such as the DSMC or the lattice Boltzmann methods (LBM).

2.2.4.1. Burnett equations

Higher-order fluid dynamic models,
also called
extended hydrodynamic equations (EHE),
may be obtained from
the conservation equations (2.39)–(2.41),

(conservation of mass, momentum and energy - LK)

deriving the form of

the viscous stress tensor σ_{EQ} and of the heat flux vector $\mathbf{q}_{\text{EQ}}$
from a molecular approach and the Boltzmann equation.

In a molecular description with the assumption of molecular chaos, a flow is entirely described if, for each time t and each position $\mathbf{x}$, we know the number density n , the velocity and the internal energy distribution functions (Lengrand and Elizarova, 2004). For example, if we consider a monatomic gas, the velocity distribution function $f(t, \mathbf{x}, \mathbf{u})$ represents the number of particles in the six-dimensional phase space $d\mathbf{x} d\mathbf{u}$ at time t . It verifies the Boltzmann equation

$$\frac{\partial f}{\partial t} + \mathbf{u} \cdot \frac{\partial f}{\partial \mathbf{x}} + \mathbf{F} \cdot \frac{\partial f}{\partial \mathbf{u}} = Q(f, f^*) \quad (2.68)$$

which is valid in the entire Knudsen regime, that is for $0 \leq \text{Kn} < \infty$, since it does not require a local thermodynamic equilibrium hypothesis. In Eq. (2.68), $\mathbf{F}$ is an external body force per unit mass and $Q(f, f^*)$ represents the intermolecular collisions. From a Chapman–Enskog expansion of the Boltzmann equation with the Knudsen number as a small parameter (i.e. $f = f_0 + \text{Kn} f_1 + \text{Kn}^2 f_2 + \dots$), the form of the viscous stress tensor and of the heat flux vector may be obtained as follows:

$$\sigma_{\text{EQ}} = \sigma_{\text{EQ}}^{(0)} + \sigma_{\text{EQ}}^{(1)} + \sigma_{\text{EQ}}^{(2)} + \sigma_{\text{EQ}}^{(3)} + \dots + \sigma_{\text{EQ}}^{(i)} + \mathcal{O}(\text{Kn}^{i+1}) \quad (2.69)$$

$$\mathbf{q}_{\text{EQ}} = \mathbf{q}_{\text{EQ}}^{(0)} + \mathbf{q}_{\text{EQ}}^{(1)} + \mathbf{q}_{\text{EQ}}^{(2)} + \mathbf{q}_{\text{EQ}}^{(3)} + \dots + \mathbf{q}_{\text{EQ}}^{(i)} + \mathcal{O}(\text{Kn}^{i+1}) \quad (2.70)$$

In the EHE, the mass flux vector $\mathbf{J}_{\text{EQ}} = \mathbf{J}_{\text{NS}} = \rho \mathbf{u}$ is the same as for the Navier–Stokes equations, but according to the number of terms kept in Eqs. (2.69) and (2.70), we obtain the Euler, Navier–Stokes, Burnett or super-Burnett equations.

Keeping only the first term in Eqs. (2.69) and (2.70), we obtain the zeroth-order approximation ($i=0$) which results in the Euler (E) equations, with

$$\sigma_E = \sigma_{EQ}^{(0)} = 0 \quad (2.71)$$

$$q_E = q_{EQ}^{(0)} = 0 \quad (2.72)$$

The **Euler equations** correspond to an **inviscid flow** ($\mu=0$) for which heat losses by thermal diffusion are neglected ($k=0$). This system of equations is a **hyperbolic system** that can describe subsonic or supersonic flows, with the transition for $Ma=1$ being modeled by a shock wave, which represents a discontinuity of the flow variables. The **Euler equations** are widely used for **high speed flows in macrosystems** such as **nozzles** and **ejectors** (Anderson, 1990).

With the **second terms** in Eqs. (2.69) and (2.70), that is for a **first-order** approximation, we obtain the **compressible NS** equations, with

$$\sigma_{NS} = \sigma_{EQ}^{(0)} + \sigma_{EQ}^{(1)} = 0 + \mu \left[\nabla \otimes \mathbf{u} + (\nabla \otimes \mathbf{u})^T - \frac{2}{3}(\nabla \cdot \mathbf{u})\mathbf{I} \right] \quad (2.73)$$

$$q_{NS} = q_{EQ}^{(0)} + q_{EQ}^{(1)} = 0 - k \nabla T \quad (2.74)$$

These classic equations are detailed in Section 2.2.2.1. According to the **associated boundary conditions**, they can be **valid** up to **Knudsen numbers in the order of 10^{-1} or 1** (cf. Fig. 2.7). In the case of a plane flow in the (x, y) plane, Eq. (2.73) yields

$$\begin{aligned} \sigma_{xx}^{(1)} &= \mu \left(\delta_1 \frac{\partial u_x}{\partial x} + \delta_2 \frac{\partial u_y}{\partial y} \right) \\ \sigma_{xy}^{(1)} &= \sigma_{yx}^{(1)} = \mu \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) \\ \sigma_{yy}^{(1)} &= \mu \left(\delta_1 \frac{\partial u_y}{\partial y} + \delta_2 \frac{\partial u_x}{\partial x} \right) \end{aligned} \quad (2.75)$$

with $\delta_1=4/3$ and $\delta_2=-2/3$.

As the Knudsen number becomes higher, additional higher-order terms from Eqs. (2.69) and (2.70) are required. The **Burnett (B) equations** correspond to a **second-order** approximation, with

$$\sigma_B = \sigma_{EQ}^{(0)} + \sigma_{EQ}^{(1)} + \sigma_{EQ}^{(2)} \quad (2.76)$$

$$q_B = q_{EQ}^{(0)} + q_{EQ}^{(1)} + q_{EQ}^{(2)} \quad (2.77)$$

These new expressions for stress and heat-flux terms dramatically complicate the EHE systems. Yun *et al.* (1998) provide the expressions of these terms in the case of a plane flow:

$$\begin{aligned}\sigma_{xx}^{(2)} = \frac{\mu^2}{p} & \left[\alpha_1 \left(\frac{\partial u_x}{\partial x} \right)^2 + \alpha_2 \frac{\partial u_x}{\partial x} \frac{\partial u_y}{\partial y} + \alpha_3 \left(\frac{\partial u_y}{\partial y} \right)^2 + \alpha_4 \frac{\partial u_x}{\partial y} \frac{\partial u_y}{\partial x} + \alpha_5 \left(\frac{\partial u_x}{\partial y} \right)^2 \right. \\ & + \alpha_6 \left(\frac{\partial u_y}{\partial x} \right)^2 + \alpha_7 R \frac{\partial^2 T}{\partial x^2} + \alpha_8 R \frac{\partial^2 T}{\partial y^2} + \alpha_9 \frac{RT}{\rho} \frac{\partial^2 \rho}{\partial x^2} + \alpha_{10} \frac{RT}{\rho} \frac{\partial^2 \rho}{\partial y^2} \\ & + \alpha_{11} \frac{RT}{\rho^2} \left(\frac{\partial \rho}{\partial x} \right)^2 + \alpha_{12} \frac{R}{\rho} \frac{\partial T}{\partial x} \frac{\partial \rho}{\partial x} + \alpha_{13} \frac{R}{T} \left(\frac{\partial T}{\partial x} \right)^2 + \alpha_{14} \frac{RT}{\rho^2} \left(\frac{\partial \rho}{\partial y} \right)^2 \\ & \left. + \alpha_{15} \frac{R}{\rho} \frac{\partial T}{\partial y} \frac{\partial \rho}{\partial y} + \alpha_{16} \frac{R}{T} \left(\frac{\partial T}{\partial y} \right)^2 \right]\end{aligned}$$

$$\begin{aligned}\sigma_{yy}^{(2)} = \frac{\mu^2}{p} & \left[\alpha_1 \left(\frac{\partial u_y}{\partial y} \right)^2 + \alpha_2 \frac{\partial u_x}{\partial x} \frac{\partial u_y}{\partial y} + \alpha_3 \left(\frac{\partial u_x}{\partial x} \right)^2 + \alpha_4 \frac{\partial u_x}{\partial y} \frac{\partial u_y}{\partial x} + \alpha_5 \left(\frac{\partial u_y}{\partial x} \right)^2 \right. \\ & + \alpha_6 \left(\frac{\partial u_x}{\partial y} \right)^2 + \alpha_7 R \frac{\partial^2 T}{\partial y^2} + \alpha_8 R \frac{\partial^2 T}{\partial x^2} + \alpha_9 \frac{RT}{\rho} \frac{\partial^2 \rho}{\partial y^2} + \alpha_{10} \frac{RT}{\rho} \frac{\partial^2 \rho}{\partial x^2} \\ & + \alpha_{11} \frac{RT}{\rho^2} \left(\frac{\partial \rho}{\partial y} \right)^2 + \alpha_{12} \frac{R}{\rho} \frac{\partial T}{\partial y} \frac{\partial \rho}{\partial y} + \alpha_{13} \frac{R}{T} \left(\frac{\partial T}{\partial y} \right)^2 + \alpha_{14} \frac{RT}{\rho^2} \left(\frac{\partial \rho}{\partial x} \right)^2 \\ & \left. + \alpha_{15} \frac{R}{\rho} \frac{\partial T}{\partial x} \frac{\partial \rho}{\partial x} + \alpha_{16} \frac{R}{T} \left(\frac{\partial T}{\partial x} \right)^2 \right] \quad (2.78)\end{aligned}$$

$$\begin{aligned}\sigma_{xy}^{(2)} = \sigma_{yx}^{(2)} = \frac{\mu^2}{p} & \left[\beta_1 \left(\frac{\partial u_x}{\partial x} \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \frac{\partial u_y}{\partial y} \right) + \beta_2 \left(\frac{\partial u_x}{\partial y} \frac{\partial u_y}{\partial y} + \frac{\partial u_x}{\partial x} \frac{\partial u_y}{\partial x} \right) \right. \\ & + \beta_3 R \frac{\partial^2 T}{\partial x \partial y} + \beta_4 \frac{RT}{\rho} \frac{\partial^2 \rho}{\partial x \partial y} + \beta_5 \frac{R}{T} \frac{\partial T}{\partial x} \frac{\partial T}{\partial y} + \beta_6 \frac{RT}{\rho^2} \frac{\partial \rho}{\partial x} \frac{\partial \rho}{\partial y} \\ & \left. + \beta_7 \frac{R}{\rho} \left(\frac{\partial T}{\partial x} \frac{\partial \rho}{\partial y} + \frac{\partial T}{\partial y} \frac{\partial \rho}{\partial x} \right) \right]\end{aligned}$$

$$\begin{aligned}
q_x^{(2)} &= \frac{\mu^2}{\rho} \left[\gamma_1 \frac{1}{T} \frac{\partial T}{\partial x} \frac{\partial u_x}{\partial x} + \gamma_2 \frac{1}{T} \frac{\partial T}{\partial x} \frac{\partial u_y}{\partial y} + \gamma_3 \frac{\partial^2 u_x}{\partial x^2} + \gamma_4 \frac{\partial^2 u_x}{\partial y^2} + \gamma_5 \frac{\partial^2 u_y}{\partial x \partial y} + \gamma_6 \frac{1}{T} \frac{\partial T}{\partial y} \frac{\partial u_y}{\partial x} \right. \\
&\quad \left. + \gamma_7 \frac{1}{T} \frac{\partial T}{\partial y} \frac{\partial u_x}{\partial y} + \gamma_8 \frac{1}{\rho} \frac{\partial \rho}{\partial x} \frac{\partial u_x}{\partial x} + \gamma_9 \frac{1}{\rho} \frac{\partial \rho}{\partial x} \frac{\partial u_y}{\partial y} + \gamma_{10} \frac{1}{\rho} \frac{\partial \rho}{\partial y} \frac{\partial u_x}{\partial y} + \gamma_{11} \frac{1}{\rho} \frac{\partial \rho}{\partial y} \frac{\partial u_y}{\partial x} \right] \\
q_y^{(2)} &= \frac{\mu^2}{\rho} \left[\gamma_1 \frac{1}{T} \frac{\partial T}{\partial y} \frac{\partial u_y}{\partial y} + \gamma_2 \frac{1}{T} \frac{\partial T}{\partial y} \frac{\partial u_x}{\partial x} + \gamma_3 \frac{\partial^2 u_y}{\partial y^2} + \gamma_4 \frac{\partial^2 u_y}{\partial x^2} + \gamma_5 \frac{\partial^2 u_x}{\partial x \partial y} + \gamma_6 \frac{1}{T} \frac{\partial T}{\partial x} \frac{\partial u_x}{\partial y} \right. \\
&\quad \left. + \gamma_7 \frac{1}{T} \frac{\partial T}{\partial x} \frac{\partial u_y}{\partial x} + \gamma_8 \frac{1}{\rho} \frac{\partial \rho}{\partial y} \frac{\partial u_y}{\partial y} + \gamma_9 \frac{1}{\rho} \frac{\partial \rho}{\partial y} \frac{\partial u_x}{\partial x} + \gamma_{10} \frac{1}{\rho} \frac{\partial \rho}{\partial x} \frac{\partial u_x}{\partial x} + \gamma_{11} \frac{1}{\rho} \frac{\partial \rho}{\partial x} \frac{\partial u_y}{\partial y} \right]
\end{aligned} \tag{2.79}$$

Unfortunately, these Burnett equations experience stability problems for very fine computational grids.

Zhong proposed in 1991 to add linear third-order terms from the super-Burnett equations (which corresponds to $i=3$) in order to stabilize the conventional Burnett equations and maintain second-order accuracy for the stress and heat flux terms (Agarwal *et al.*, 2001). This set of equations is called the augmented Burnett (AB) equations.

Moreover, Welder *et al.* (1993) have shown that a linear stability analysis is not sufficient to explain the instability of the Burnett equations. This instability could be due to the fact that the conventional Burnett equations can violate the second law of thermodynamics at high Knudsen numbers (Comeaux *et al.*, 1995). To overcome this difficulty, Balakrishnan and Agarwal have proposed another set of equations termed the Bhatnagar–Gross–Krook–Burnett or BGK–Burnett (BGKB) equations, which are consistent with the second principle and unconditionally stable for both monatomic and polyatomic gases. The BGKB equations also use third-order terms but the second-order terms $\{\alpha_1, \dots, \alpha_{16}, \beta_1, \dots, \beta_7, \gamma_1, \dots, \gamma_{11}\}$ are different than the ones of the conventional Burnett or AB equations, as well as the first-order terms ($\delta_1=-1.6$ and $\delta_2=-0.4$ for $\gamma=1.4$). The additional terms for the AB equations are

$$\begin{aligned}
\sigma_{xx}^{(a)} &= \frac{\mu^3}{p^2} RT \left[\alpha_{17} \left(\frac{\partial^3 u_x}{\partial x^3} + \frac{\partial^3 u_x}{\partial x \partial y^2} \right) + \alpha_{18} \left(\frac{\partial^3 u_y}{\partial y^3} + \frac{\partial^3 u_y}{\partial y \partial x^2} \right) \right] \\
\sigma_{yy}^{(a)} &= \frac{\mu^3}{p^2} RT \left[\alpha_{17} \left(\frac{\partial^3 u_y}{\partial y^3} + \frac{\partial^3 u_y}{\partial y \partial x^2} \right) + \alpha_{18} \left(\frac{\partial^3 u_x}{\partial x^3} + \frac{\partial^3 u_x}{\partial x \partial y^2} \right) \right] \\
\sigma_{xy}^{(a)} &= \sigma_{yx}^{(a)} = \frac{\mu^3}{p^2} RT \beta_8 \left(\frac{\partial^3 u_x}{\partial y^3} + \frac{\partial^3 u_y}{\partial x^3} + \frac{\partial^3 u_x}{\partial y \partial x^2} + \frac{\partial^3 u_y}{\partial x \partial y^2} \right)
\end{aligned} \tag{2.80}$$

$$\begin{aligned}
q_x^{(a)} &= \frac{\mu^3}{p\rho} R \left[\gamma_{12} \left(\frac{\partial^3 T}{\partial x^3} + \frac{\partial^3 T}{\partial x \partial y^2} \right) + \gamma_{13} \frac{T}{\rho} \left(\frac{\partial^3 \rho}{\partial x^3} + \frac{\partial^3 \rho}{\partial x \partial y^2} \right) \right] \\
q_y^{(a)} &= \frac{\mu^3}{p\rho} R \left[\gamma_{12} \left(\frac{\partial^3 T}{\partial y^3} + \frac{\partial^3 T}{\partial y \partial x^2} \right) + \gamma_{13} \frac{T}{\rho} \left(\frac{\partial^3 \rho}{\partial y^3} + \frac{\partial^3 \rho}{\partial y \partial x^2} \right) \right]
\end{aligned} \tag{2.81}$$

and for the BGK–Burnett equations:

$$\begin{aligned}
\sigma_{xx}^{(B)} &= \frac{\mu^3}{p^2} RT \left[\theta_1 \frac{\partial^3 u_x}{\partial x^3} + \theta_2 \frac{\partial^3 u_x}{\partial x \partial y^2} + \theta_3 \frac{\partial^3 u_y}{\partial x^2 \partial y} + \theta_4 \frac{\partial^3 u_y}{\partial y^3} \right] \\
&\quad - \frac{\mu^3}{p^2} \frac{RT}{\rho} \left[\frac{\partial \rho}{\partial x} \left(\theta_1 \frac{\partial^2 u_x}{\partial x^2} + \theta_5 \frac{\partial^2 u_y}{\partial x \partial y} + \theta_6 \frac{\partial^2 u_x}{\partial y^2} \right) \right. \\
&\quad \quad \left. + \frac{\partial \rho}{\partial y} \left(\theta_7 \frac{\partial^2 u_y}{\partial x^2} + \theta_8 \frac{\partial^2 u_x}{\partial x \partial y} + \theta_4 \frac{\partial^2 u_y}{\partial y^2} \right) \right] \\
&\quad + \frac{\mu^3}{p^2} \left[\theta_9 \left(\frac{\partial u_x}{\partial x} \right)^3 + \theta_{10} \left\{ 3 \left(\frac{\partial u_x}{\partial x} \right)^2 \frac{\partial u_y}{\partial y} + \left(\frac{\partial u_y}{\partial y} \right)^3 \right\} + \theta_{11} \frac{\partial u_x}{\partial x} \left(\frac{\partial u_y}{\partial y} \right)^2 \right. \\
&\quad \quad \left. - \left(\theta_4 \frac{\partial u_x}{\partial x} + \theta_{12} \frac{\partial u_y}{\partial y} \right) \left\{ \left(\frac{\partial u_x}{\partial y} \right)^2 + 2 \frac{\partial u_x}{\partial y} \frac{\partial u_y}{\partial x} + \left(\frac{\partial u_y}{\partial x} \right)^2 \right\} \right] \\
&\quad + \frac{\mu^3}{p^2} R \left[\left(\theta_{13} \frac{\partial u_x}{\partial x} + \theta_{14} \frac{\partial u_y}{\partial y} \right) \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \right] \\
\sigma_{yy}^{(B)} &= \frac{\mu^3}{p^2} RT \left[\theta_1 \frac{\partial^3 u_y}{\partial y^3} + \theta_2 \frac{\partial^3 u_y}{\partial y \partial x^2} + \theta_3 \frac{\partial^3 u_x}{\partial y^2 \partial x} + \theta_4 \frac{\partial^3 u_x}{\partial x^3} \right] \\
&\quad - \frac{\mu^3}{p^2} \frac{RT}{\rho} \left[\frac{\partial \rho}{\partial y} \left(\theta_1 \frac{\partial^2 u_y}{\partial y^2} + \theta_5 \frac{\partial^2 u_x}{\partial x \partial y} + \theta_6 \frac{\partial^2 u_y}{\partial x^2} \right) \right. \\
&\quad \quad \left. + \frac{\partial \rho}{\partial x} \left(\theta_7 \frac{\partial^2 u_x}{\partial y^2} + \theta_8 \frac{\partial^2 u_y}{\partial x \partial y} + \theta_4 \frac{\partial^2 u_x}{\partial x^2} \right) \right] \\
&\quad + \frac{\mu^3}{p^2} \left[\theta_9 \left(\frac{\partial u_y}{\partial y} \right)^3 + \theta_{10} \left\{ 3 \left(\frac{\partial u_y}{\partial y} \right)^2 \frac{\partial u_x}{\partial x} + \left(\frac{\partial u_x}{\partial x} \right)^3 \right\} + \theta_{11} \frac{\partial u_y}{\partial y} \left(\frac{\partial u_x}{\partial x} \right)^2 \right. \\
&\quad \quad \left. - \left(\theta_4 \frac{\partial u_y}{\partial y} + \theta_{12} \frac{\partial u_x}{\partial x} \right) \left\{ \left(\frac{\partial u_y}{\partial x} \right)^2 + 2 \frac{\partial u_x}{\partial y} \frac{\partial u_y}{\partial x} + \left(\frac{\partial u_x}{\partial y} \right)^2 \right\} \right] \\
&\quad + \frac{\mu^3}{p^2} R \left[\left(\theta_{13} \frac{\partial u_y}{\partial y} + \theta_{14} \frac{\partial u_x}{\partial x} \right) \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \right]
\end{aligned} \tag{2.82}$$

$$\begin{aligned}
\sigma_{xy}^{(B)} = & \frac{\mu^3}{p^2} RT \left[\theta_{15} \left(\frac{\partial^3 u_x}{\partial y \partial x^2} + \frac{\partial^3 u_y}{\partial y^2 \partial x} \right) + \frac{\partial^3 u_x}{\partial y^3} + \frac{\partial^3 u_y}{\partial x^3} \right] \\
& - \frac{\mu^3}{p^2} \frac{RT}{\rho} \left[\frac{\partial \rho}{\partial y} \left(\theta_6 \frac{\partial^2 u_x}{\partial x^2} + \theta_{16} \frac{\partial^2 u_y}{\partial x \partial y} + \frac{\partial^2 u_x}{\partial y^2} \right) + \frac{\partial \rho}{\partial x} \left(\frac{\partial^2 u_y}{\partial x^2} + \theta_{16} \frac{\partial^2 u_x}{\partial x \partial y} + \theta_6 \frac{\partial^2 u_y}{\partial y^2} \right) \right] \\
& - \frac{\mu^3}{p^2} \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) \left[\theta_4 \left\{ \left(\frac{\partial u_x}{\partial x} \right)^2 + \left(\frac{\partial u_y}{\partial y} \right)^2 \right\} + 2\theta_{12} \frac{\partial u_x}{\partial x} \frac{\partial u_y}{\partial y} + \theta_7 \left\{ \left(\frac{\partial u_y}{\partial x} \right)^2 \right. \right. \\
& \left. \left. + 2 \frac{\partial u_x}{\partial y} \frac{\partial u_y}{\partial x} + \left(\frac{\partial u_x}{\partial y} \right)^2 \right\} \right] + \frac{\mu^3}{p^2} R \left[\theta_{17} \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \right] \\
q_x^{(B)} = & \frac{\mu^3}{p\rho} R \theta_{18} \left[\frac{\partial^3 T}{\partial x^3} + \frac{\partial^3 T}{\partial y^2 \partial x} - \frac{1}{\rho} \frac{\partial \rho}{\partial x} \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \right] \\
& + \frac{\mu^3}{p\rho} \left[\frac{\partial u_x}{\partial x} \left(\theta_{19} \frac{\partial^2 u_x}{\partial x^2} + \theta_{20} \frac{\partial^2 u_y}{\partial x \partial y} + \theta_6 \frac{\partial^2 u_x}{\partial y^2} \right) \right. \\
& \left. + \frac{\partial u_y}{\partial y} \left(\theta_{21} \frac{\partial^2 u_x}{\partial x^2} + \theta_{22} \frac{\partial^2 u_y}{\partial x \partial y} + \theta_7 \frac{\partial^2 u_x}{\partial y^2} \right) \right. \\
& \left. + \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) \left(\theta_{23} \frac{\partial^2 u_y}{\partial x^2} + \theta_{24} \frac{\partial^2 u_x}{\partial x \partial y} + \theta_6 \frac{\partial^2 u_y}{\partial y^2} \right) \right] \\
& - \frac{\mu^3}{p\rho} \left(\frac{1}{\rho} \frac{\partial \rho}{\partial x} + \frac{1}{T} \frac{\partial T}{\partial x} \right) \left[\theta_{13} \left\{ \left(\frac{\partial u_x}{\partial x} \right)^2 + \left(\frac{\partial u_y}{\partial y} \right)^2 \right\} \right. \\
& \left. + 2\theta_{14} \frac{\partial u_x}{\partial x} \frac{\partial u_y}{\partial y} + \theta_{17} \left\{ 2 \frac{\partial u_x}{\partial y} \frac{\partial u_y}{\partial x} + \left(\frac{\partial u_x}{\partial y} \right)^2 + \left(\frac{\partial u_y}{\partial x} \right)^2 \right\} \right] \\
& + \frac{\mu^3}{p\rho} \frac{R}{T} \frac{\partial T}{\partial x} \theta_{18} \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)
\end{aligned} \tag{2.83}$$

$$\begin{aligned}
q_y^{(B)} = & \frac{\mu^3}{p\rho} R\theta_{18} \left[\frac{\partial^3 T}{\partial y^3} \frac{\partial^3 T}{\partial x^2 \partial y} - \frac{1}{\rho} \frac{\partial \rho}{\partial y} \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \right] \\
& + \frac{\mu^3}{p\rho} \left[\frac{\partial u_y}{\partial y} \left(\theta_{19} \frac{\partial^2 u_y}{\partial y^2} + \theta_{20} \frac{\partial^2 u_x}{\partial x \partial y} + \theta_6 \frac{\partial^2 u_y}{\partial x^2} \right) \right. \\
& \quad + \frac{\partial u_x}{\partial x} \left(\theta_{21} \frac{\partial^2 u_y}{\partial y^2} + \theta_{22} \frac{\partial^2 u_x}{\partial x \partial y} + \theta_7 \frac{\partial^2 u_y}{\partial x^2} \right) \\
& \quad \left. + \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) \left(\theta_{23} \frac{\partial^2 u_x}{\partial y^2} + \theta_{24} \frac{\partial^2 u_y}{\partial x \partial y} + \theta_6 \frac{\partial^2 u_x}{\partial x^2} \right) \right] \\
& - \frac{\mu^3}{p\rho} \left(\frac{1}{\rho} \frac{\partial \rho}{\partial y} + \frac{1}{T} \frac{\partial T}{\partial y} \right) \left[\theta_{13} \left\{ \left(\frac{\partial u_x}{\partial x} \right)^2 + \left(\frac{\partial u_y}{\partial y} \right)^2 \right\} \right. \\
& \quad \left. + 2\theta_{14} \frac{\partial u_x}{\partial x} \frac{\partial u_y}{\partial y} + \theta_{17} \left\{ 2 \frac{\partial u_x}{\partial y} \frac{\partial u_y}{\partial x} + \left(\frac{\partial u_x}{\partial y} \right)^2 + \left(\frac{\partial u_y}{\partial x} \right)^2 \right\} \right] \\
& + \frac{\mu^3}{p\rho} \frac{R}{T} \frac{\partial T}{\partial y} \theta_{18} \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)
\end{aligned}$$

The different coefficients for the above equations are given in [Table 2.4](#).

Table 2.4

Coefficients for AB and BGKB equations (from Yun *et al.* (1998)); the coefficients for AB equations correspond to an HS collision model, and the coefficients for BGKB equations correspond to $\gamma = 1.4$.

	α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8	α_9	α_{10}
AB	1.199	0.153	-0.600	-0.115	1.295	-0.733	0.260	-0.130	-1.352	0.676
BGKB	-2.24	-0.48	0.56	-1.2	0.0	0.0	-19.6	-5.6	-1.6	0.4
	α_{11}	α_{12}	α_{13}	α_{14}	α_{15}	α_{16}	α_{17}	α_{18}	β_1	β_2
AB	1.352	-0.898	0.600	-0.676	0.449	-0.300	0.2222	-0.1111	-0.115	1.913
BGKB	1.6	-19.6	-18.0	-0.4	-5.6	-6.9	-	-	-1.4	-1.4
	β_3	β_4	β_5	β_6	β_7	β_8	γ_1	γ_2	γ_3	γ_4
AB	0.390	-2.028	0.900	2.028	-0.676	0.1667	10.830	0.407	-2.269	1.209
BGKB	0.0	-2.0	2.0	2.0	0.0	-	-25.241	-0.2	-1.071	-2.0
	γ_5	γ_6	γ_7	γ_8	γ_9	γ_{10}	γ_{11}	γ_{12}	γ_{13}	
AB	-3.478	-0.611	11.033	-2.060	1.030	-1.545	-1.545	0.6875	-0.625	
BGKB	-2.8	-7.5	-11.0	-1.271	1.0	-3.0	-3.0	-	-	
	θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	θ_7	θ_8	θ_9	θ_{10}
BGKB	2.56	1.36	0.56	-0.64	0.96	1.6	-0.4	-0.24	1.024	-0.256
	θ_{11}	θ_{12}	θ_{13}	θ_{14}	θ_{15}	θ_{16}	θ_{17}	θ_{18}	θ_{19}	θ_{20}
BGKB	1.152	0.16	2.24	-0.56	3.6	0.6	1.4	4.9	7.04	-0.16
	θ_{21}	θ_{22}	θ_{23}	θ_{24}	δ_1	δ_2				
BGKB	-1.76	4.24	3.8	3.4	-1.6	-0.4				

2.2.4.2. DSMC method

For high Knudsen numbers, the continuum approach (NS, QGD, QHD or B equations) is no longer valid, even with slip boundary conditions.

A molecular approach is then required and the flow may be described from the Boltzmann equation (2.68).

However, the integral formulation of $Q(f, f^*)$, which represents the binary intermolecular collisions, is complex in the general case due to non-linearities and a great number of independent variables (Karniadakis and Beskok, 2002). Therefore, the numerical resolution of the Boltzmann equation is reserved to problems with a simple geometry or problems for which the rarefaction level allows simplifications. Thus, $Q(f, f^*)$ is zero in the free molecular regime when $Kn \rightarrow \infty$. Also, when $Kn \rightarrow 0$, a semi-analytical resolution of the Boltzmann equation can be obtained using the method of moments of Grad, for which the distribution function is represented by a series of orthogonal Hermite polynomials (Grad, 1949) or the Chapman–Enskog method (Chapman and Cowling, 1952), for which the distribution function is expanded into a perturbation series with the Knudsen number being the small parameter (*cf.* Section 2.2.4.1). In the transition regime on the other hand, the numerical resolution of the Boltzmann equation requires approximate methods based on the simplification of $Q(f, f^*)$. Sharipov and Selenev (1998) describe the different available methods – BGK equation (Bhatnagar *et al.*, 1954), linearized Boltzmann equation (Cercignani *et al.*, 1994)... – with their conditions of validity.

The Direct Simulation Monte Carlo (DSMC) method is actually better suited to the transition regime. This method was developed by Bird (1978; 1998). Initially widely used for the simulation of low-pressure rarefied flows (Muntz, 1989; Cheng and Emmanuel, 1995), it is now often used for microfluidic applications (Stefanov and Cercignani, 1994; Piekos and Breuer, 1996; Mavriplis *et al.*, 1997; Chen *et al.*, 1998; Oran *et al.*, 1998; Hudson and Bartel, 1999; Pan *et al.*, 1999; Pan *et al.*, 2000; Pan *et al.*, 2001; Wu and Tseng, 2001; Wang and Li, 2004).

As for the classical kinetic theory, the DSMC method assumes a molecular chaos and a dilute gas. The technique is equivalent to solving the Boltzmann equation for a monatomic gas with binary collisions. The DSMC technique consists in spitting off the simulations of the intermolecular collisions and of the molecular motion, with a time step smaller than the mean collision time $1/\nu$. The simulation is performed using a limited number of molecules, with each simulated molecule representing a great number W of real molecules. The

control volume is divided into cells, whose size is of the order of $\lambda/3$, in order to correctly treat large gradient regions. The simulation involves four main steps (Oran *et al.*, 1998):

- (1) Moving simulated molecules and modeling molecule–surface interactions, applying conservation laws.
- (2) Indexing and tracking molecules.
- (3) Simulating collisions with a probabilistic process, generally with a VHS model.
- (4) Sampling the macroscopic flow properties at the geometric center of the cells.

The statistical error of the DSMC solution is inversely proportional to the square root of the total number of simulated molecules. The technique being explicit and time-marching, the simulation is unsteady. The solution of an unsteady problem is obtained from averaging an ensemble of many independent computations, to reduce statistical errors. For a steady problem, each independent computation is proceeded until a steady flow is established (Fig. 2.9).

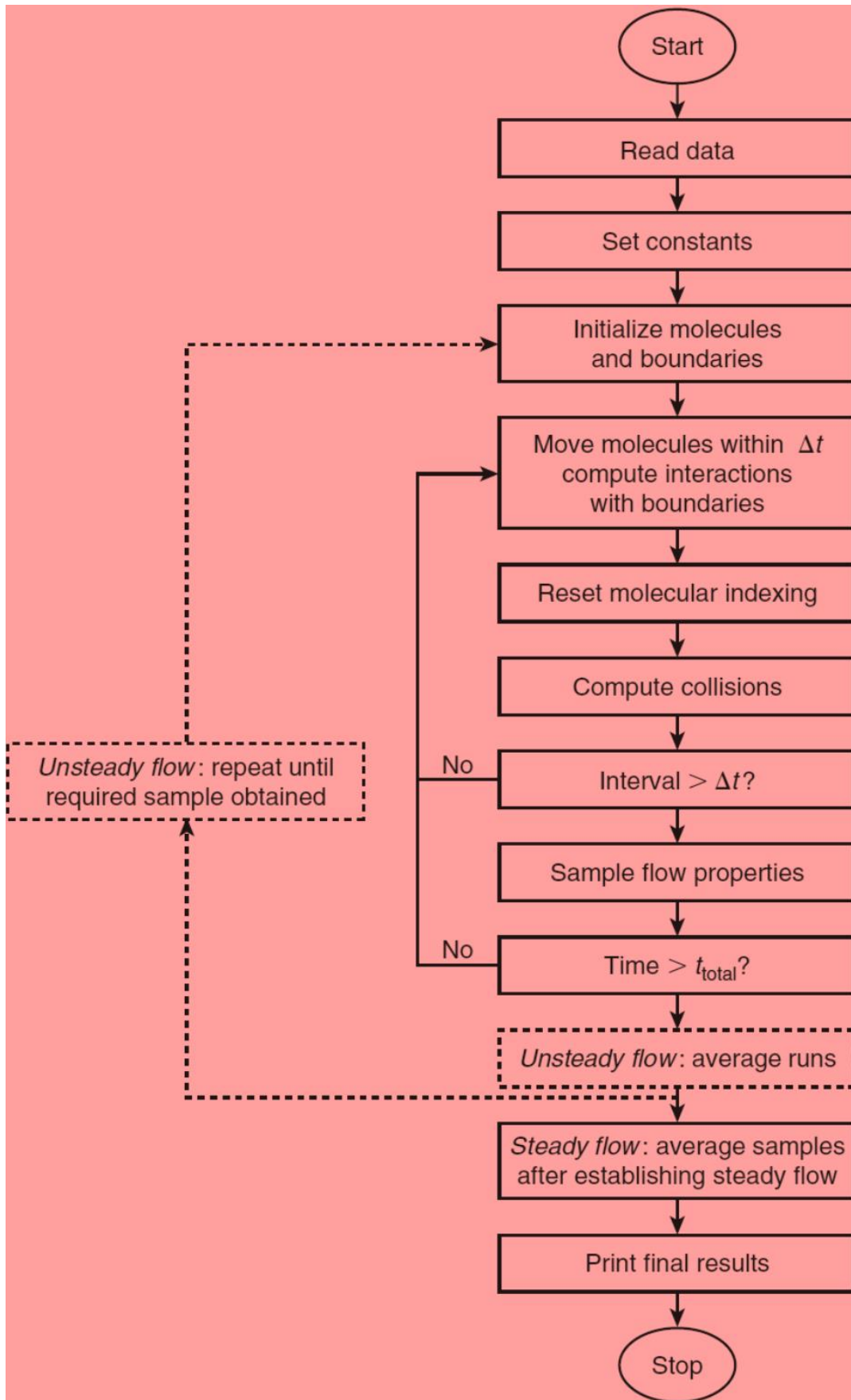


Fig. 2.9. DSMC flowchart (from Oran *et al.* (1998)).

2.2.4.3. Lattice Boltzmann method

The lattice Boltzmann method (LBM) is appropriate for complex geometries and seems particularly effective for the simulation of flows in microsystems, where both mesoscopic dynamics and microscopic statistics are important (Karniadakis and Beskok, 2002). The method, whose detailed description is proposed by Chen and Doolen (1998), solves a simplified Boltzmann equation on a discrete lattice. The use of LBM for the simulation of gas microflows is recent, but LBM is destined to be more widely used for such applications in the future. For example, Nie *et al.* (2002) demonstrated that the LBM can capture the fundamental behaviors in microchannel flows, including velocity slip, nonlinear pressure drop along the channel and mass flowrate variation with Knudsen number. The numerical results found by Lim *et al.* (2002) are also in good agreement with the data obtained analytically and experimentally for a two-dimensional isothermal pressure-driven microchannel flow.